Kaggle

Supply Chain Data Overview

Key Findings and Recommendations

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Insights & Strategic Analysis

- PLANT16 has the highest unit cost ($1.91). Small volume shifts to lower-cost plants like PLANT07 ($0.37) could result in significant savings.
- Carrier V44_3 shows inconsistent delivery times. Shifting high-priority orders to more reliable partner
- Analysis of WhCapacities shows specific plants are approaching their daily limits during peak months. Increase temporary labor at high-volume ports.
- Our math shows that the number of items ship affects the cost more than how heavy they are. To save money, you should combine many small packages into one big shipment

Dashboard Design & Visualization Strategy

- High-level KPI cards for Volume, Cost, and OTD%, paired with a Map visual to show geographic distribution of Origin Ports.
- A Bar Chart comparing average delays by Carrier and a Treemap analyzing Total Cost by Plant Code.
- Includes interactive Slicers (Year, Plant, Carrier) and a Line Chart showing YoY Volume Change to identify growth patterns

DAX Calculations & Measures

Core DAX Metrics Overview

- Total Volume: = **SUM(Unit Quantity)**
- Total WH Cost: = **SUMX(Unit Quantity * Cost/unit)**
- On-Time Delivery %: = **On-Time Orders ÷ Total Orders**
- Avg. Cost per Unit: = **Total WH Cost ÷ Total Volume**

Descriptive Statistical Analysis

Key Findings and Insights

- Mean Order Volume: **3,202.75 units**
- Weight Variability: Std. Dev. **66.57**
- Correlation of Weight vs. Ship Late Day Count: **-0.017**
- Indicates **no linear relationship** suggesting delays due to systemic bottlenecks.
- Regression analysis shows that Unit Quantity is a significant predictor of Total Warehouse Cost.

Data Cleaning & Transformation

Ensuring Analysis Reliability

- Managed the Ship Late Day count column by filling nulls with 0, ensuring that "no data" was correctly treated as "no delay."
- Used Power Query to convert Order Date into a standard Date format and ensured Unit Quantity and Weight are Decimal Numbers for mathematical accuracy.
- Created the On-Time Status column (Logic: If Late Days = 0, then "On-Time"; else "Late").